RATIONAL HOMOTOPY THEORY – ASSIGNMENT 1,2,3,4

- **Below, there are four assignments that you have to submit before mid-semester. Read the questions carefully.**
- **Questions, e.g., 1, 2, 3, etc., are from Hatcher's algebraic topology, so go through the book.**
- **Send me an email in case you have any questions.**

1. Assignment 1 : Due on 22nd January 2026

(1) Compute directly the simplicial homology of a figure 8, and give a cellular decomposition for this space.

(2) Show that $S^n \subset \mathbb{R}^{n+1}$ is a deformation retract of $\mathbb{R}^{n+1} - \{0\}$.

(3) Consider the simplicial degree 2 map $S^1 \to S^1$ we discussed in class. Calculate its induced map on homology and on homotopy groups.

(4) (a) Consider the inclusion $S^n \xrightarrow{i} D^{n+1}$ of the n-sphere as the boundary of the $(n+1)$-disk $D^{n+1} = \{x \in \mathbb{R}^{n+1}, |x| \leq 1\}$. Prove that the disk does not retract onto the sphere, i.e. that there is no (continuous) map $D^{n+1} \to S^n$ such that the composition ri equals the identity map $S^n \to S^n$.

(b) Prove the Brouwer fixed theorem for disks: Every continuous map $D^n \xrightarrow{f} D^n$ has a fixed point, i.e. a point $x \in D^n$ such that $f(x) = x$. Hint: suppose there is no fixed point, then x and $f(x)$ are always distinct and so determine a line. Using this build a retract to the boundary. You do not have to carefully argue that your constructed map is continuous.

(c) Prove that the Brouwer fixed point theorem holds for any space homeomorphic to a disk D^n.

(d) Show that the Brouwer fixed point theorem need not hold on a space that is only homotopy equivalent to a disk. That is, find an example of a space X homotopy equivalent to a disk, together with a continuous map $X \xrightarrow{f} X$, which has no fixed points. Moral of the story: homeomorphism type knows about the individual points in a space, while the homotopy type does not.

The Brouwer fixed point theorem can be used to recover many disparate results: the fundamental theorem of algebra, the Nash equilibrium theorem, and more.

(e) Here is another basic application: prove that a matrix with non-negative real entries has at least one non-negative eigenvalue. Hint: if zero is an eigenvalue, then we're done; otherwise every non-zero vector with non-negative entries is sent to a non-zero vector with non-negative entries. You can take for granted that the intersection of the unit sphere in $\mathbb{R}^n$ with the set of vectors with non-negative entries is homeomorphic to a disk. (E.g. in $\mathbb{R}^2$ this intersection is a quarter of a circle, from 0 to $\frac{\pi}{2}$.)

(5) Consider a finite chain complex of finite-dimensional vector spaces C_i,

$$0 \xrightarrow{\partial} C_k \xrightarrow{\partial} C_{k-1} \xrightarrow{\partial} \cdots \xrightarrow{\partial} C_0 \xrightarrow{\partial} 0$$

(A finite chain complex is a chain complex with finitely many non-zero terms.) Define its **Euler characteristic** to be the alternating sum of dimensions of homology

$$\dim H_0 - \dim H_1 + \dim H_2 - \cdots + (-1)^k \dim H_k.$$

Now consider another chain complex, with the same vector spaces but another differential ∂':

$$0 \xrightarrow{\partial'} C_k \xrightarrow{\partial'} C_{k-1} \xrightarrow{\partial'} \cdots \xrightarrow{\partial'} C_0 \xrightarrow{\partial'} 0$$

- Prove that the Euler characteristics of both chain complexes coincide.

That is, the Euler characteristic depends only on the vector spaces in the chain complex and not on the differential. In particular, taking the trivial differential $\partial'' = 0$, we conclude that the Euler characteristic is equal to the alternating sum

$$\dim C_0 - \dim C_1 + \dim C_2 - \cdots + (-1)^k \dim C_k.$$

We can take C_i to be the simplicial or cellular chains of some finite simplicial complex or finite CW complex. Since simplicial and cellular homology are isomorphic, the Euler characteristics of the resulting complexes are equal; furthermore, by the above, we can compute the Euler characteristic by taking the alternating sum of the number of i-simplices or i-cells. We denote the Euler characteristic of a topological space X (computed in any of these equivalent ways) by $\chi(X)$.

- Calculate the Euler characteristic of S^2.
- Calculate the Euler characteristic of S^n, for n a non-negative integer.
- Calculate the Euler characteristic of $S^n \times S^n$.
- Extra credit: Make a guess for a general formula for the Euler characteristic of a product $\chi(X \times Y)$. Use the Künneth theorem to prove it.

(6) Prove that the Hopf fiber bundle $S^1 \to S^3 \xrightarrow{p} S^2$ does not have a section, i.e. there is no map $S^2 \xrightarrow{s} S^3$ such that the composition ps is the identity map $S^2 \to S^2$.

(7) Show that $H_2(\mathbb{RP}^2 \times S^3; \mathbb{Z}) = 0$.

(8) Let X, Y be topological spaces and let $f : X \to Y$ be a continuous map. The *mapping cone* of f is the space

$$C_f = (X \times I \sqcup Y)/\sim,$$

where $(x, 1) \sim f(x)$ for all $x \in X$, and $X \times \{0\}$ is collapsed to a point.

(a) Assume that X and Y are path-connected. Express $\pi_1(C_f)$ as a pushout involving $\pi_1(X)$, $\pi_1(Y)$, and the induced homomorphism $f_* : \pi_1(X) \to \pi_1(Y)$.

(b) Let $X = Y = S^1 \vee S^1$ with the 1–cells labeled a, b. Let $f : X \to Y$ be defined on fundamental groups by

$$f_*(a) = aba^{-1}b^{-1}, \qquad f_*(b) = a^2b^3.$$

Compute $\pi_1(C_f)$ explicitly.

(9) Let X be a topological space and let $f : X \to X$ be a continuous map. The *mapping torus* of f is the space

$$T_f = X \times I/(x, 1) \sim (f(x), 0).$$

Suppose now that X is a connected CW–complex with $x_0 \in X$ the only 0–cell, and that $f : X \to X$ is a cellular map satisfying $f(x_0) = x_0$. Then T_f is also a CW–complex whose cells are of the form $e \times \{0\}$ and $e \times (0, 1)$ for each cell e of X (you do not need to prove this).

Assume that

$$\pi_1(X, x_0) = \langle A \mid R \rangle.$$

Prove that

$$\pi_1(T_f) = \left\langle A, t \;\middle|\; R,\ t^{-1}at = f_*(a) \text{ for all } a \in A \right\rangle.$$

Do Problem 11 in Section 1.2 (of Hatcher Algebraic topology) as an example.

Hint: The generator t corresponds to the loop $\{x_0\} \times I$.

Note: The assumption that X has only one 0–cell is for convenience; the general case can be reduced to this situation.

(10) Find all connected covering spaces of $\mathbb{RP}^2 \vee \mathbb{RP}^2$.

2. Assignment 2 : Due on 29th January 2026

(1) Section 3.2, Problem 1.
(2) Section 3.2, Problem 3.
(3) Section 3.2, Problem 4.
(4) Section 3.2, Problem 7.
(5) Section 3.2, Problem 11.
(6) Section 3.2, Problem 18.
(7) Prove that the degree of any map from S^4 to $S^2 \times S^2$ is zero. (See Section 3.3, Problem 7 for the definition of degree.)
(8) Prove that there is no map $f : \mathbb{CP}^2 \to \mathbb{CP}^2$ of negative degree. **Hint:** The induced homomorphism
$$f_* : H_4(\mathbb{CP}^2; \mathbb{Z}) \to H_4(\mathbb{CP}^2; \mathbb{Z})$$
is multiplication by $\deg(f)$.
(9) Use Poincaré duality to show that $\chi(M) = 0$ for every closed connected manifold of odd dimension. (We did this for orientable manifolds in class. How do you show it for non-orientable manifolds?)
(10) Let X be a space such that
$$H_0(X; \mathbb{Z}) \cong \mathbb{Z}, \qquad H_1(X; \mathbb{Z}) \cong H_2(X; \mathbb{Z}) \cong \mathbb{Z} \oplus \mathbb{Z}/6\mathbb{Z}, \qquad H_3(X; \mathbb{Z}) \cong \mathbb{Z},$$
and
$$H_i(X; \mathbb{Z}) = 0 \quad \text{for } i > 3.$$
Compute $H_i(X; \mathbb{Z}/2\mathbb{Z})$, $H^i(X; \mathbb{Z})$, and $H^i(X; \mathbb{Z}/2\mathbb{Z})$ for all i.

3. Assignment 3 : Due on 5th February

(1) Prove that $\mathbb{CP}^\infty$ is an Eilenberg–Maclane space of type $K(\mathbb{Z}, 2)$.

(2) Define the **Betti numbers** b_i of a space X be $b_i = \dim H^i(X; \mathbb{Q})$. Prove that a path connected space X with finite fundamental group has $b_1 = 0$.
   - Extra credit: Is the converse true?

(3) Prove that the Euler characteristic of a smooth closed orientable odd-dimensional manifold is zero.

(4) Find a simply connected closed n–manifold M with $\pi_{n+1}(M) = 0$, for some positive n. You can freely use tables of known homotopy groups of spheres.

(5) Let M be a closed orientable smooth simply connected three–manifold. (The orientability hypothesis is in fact unnecessary, as it is implied by simple connectivity.) Prove that M is homotopy equivalent to S^3. Show that for every $n \geq 4$, a simply connected closed n–manifold need not be homotopy equivalent to S^n.

(6) Let p be an odd prime. Prove that every fibration over $\Sigma\mathbb{RP}^2$ with fiber $K(\mathbb{Z}/p\mathbb{Z}, 2)$ has a section.

(7) Compute the cohomology ring $H^*(\Omega S^3; \mathbb{Z})$ of the based loop space of the three–sphere. Show that there is a map $\Omega S^3 \to \mathbb{CP}^\infty$ inducing an isomorphism on rational cohomology $H^*(-; \mathbb{Q})$.

4. Assignment 4 : Due on 12th February

(a) Compute $\pi_5(S^4)$ using the Postnikov tower of S^4.

(b) Let $F \to X \to B$ be a fibration of finite CW complexes with B simply connected. Prove, using the Leray–Serre spectral sequence, that the Euler characteristic is multiplicative, i.e. $\chi(X) = \chi(B)\chi(F)$. Hint: run the spectral sequence with rational coefficients, and employ the property that the Euler characteristic of a chain complex does not depend on the differential from Assignment 1. The E_2 page contains $H^*(B;\mathbb{Q})$ and $H^*(F;\mathbb{Q})$, while the E_∞ page contains $H^*(X;\mathbb{Q})$.

(c) Extra credit: Prove that for simply connected spaces with finitely many non-zero homotopy groups, for any map $X \xrightarrow{f} Y$ there is an induced map of Postnikov towers. That is, there are maps $X_k \to Y_k$ between the corresponding stages in the Postnikov tower making the diagram

$$\begin{tikzcd}
X \arrow[r] \arrow[d] & Y \arrow[d] \\
\vdots \arrow[d] & \vdots \arrow[d] \\
X_k \arrow[r] \arrow[d] & Y_k \arrow[d] \\
X_{k-1} \arrow[r] \arrow[d] & Y_{k-1} \arrow[d] \\
\vdots \arrow[d] & \vdots \arrow[d] \\
X_2 \arrow[r] & Y_2
\end{tikzcd}$$

commutative up to homotopy. The hypothesis that there are only finitely many non-zero homotopy groups is not necessary. Can you remove it?

(d) Prove that the rationalization of $\Sigma\mathbb{CP}^2$ is homotopy equivalent to $S^3_{\mathbb{Q}} \vee S^5_{\mathbb{Q}}$. Hint: look at the last slide of module 7, and use that $\Sigma\mathbb{CP}^2$ is the mapping cone of the suspension of the Hopf map $S^4 \to S^3$. You will also want to consider the mapping cone of the induced map $S^4_{\mathbb{Q}} \to S^3_{\mathbb{Q}}$. The homotopy type of the mapping cone depends only on the homotopy class of the map; what can you say about $S^4_{\mathbb{Q}} \to S^3_{\mathbb{Q}}$?

(e) Give, with proof, an example of two $\mathbb{Q}$-cdga's such that there is no quasi-isomorphism $(A,d) \to (B,d)$, but such that there is a quasi-isomorphism $(A,d)\otimes\mathbb{R} \to (B,d)\otimes\mathbb{R}$ of $\mathbb{R}$-cdga's. (Tensoring (A,d) with $\mathbb{R}$ means taking $A\otimes\mathbb{R}$ and extending the differential d by setting $d(a\otimes r) = (da)\otimes r$, for $a\in A, r\in\mathbb{R}$. You do not have to write the $\otimes$ symbol; this is just the benign operation of extending the scalars from $\mathbb{Q}$ to $\mathbb{R}$.)

(f) The free commutative graded $\mathbb{Q}$-algebra $\Lambda(x,y,z,w)$, with $\deg(x) = \deg(y) = \deg(z) = \deg(w) = 1$ can be equipped in various ways with a differential making it into a minimal cdga. Parametrize all such differentials and classify them based on the cohomology algebra (up to isomorphism) of the resulting cdga.

(g) Prove the *rational Hurewicz theorem*: Let X be a simply connected space, with degree-wise finitely generated homology, such that $\pi_{\leq n}(X)\otimes\mathbb{Q} = 0$. Then, for $n+1 \leq i \leq 2n$, $\pi_i(X)\otimes\mathbb{Q} \cong H_i(X;\mathbb{Q})$, and $\pi_{2n+1}(X)\otimes\mathbb{Q}$ surjects onto $H_{2n+1}(X;\mathbb{Q})$. Hint: look at a minimal model of X.